2D Temperature problem (Convection and diffusion) using FVM with 1st order UDS

Roger Stan Hervet

October 2026

Introduction

This project solves the steady 2D Temperature equation using first order Upwind differencing scheme with finite volume method.We consider both convection and diffusion transport across the domain.The upwind differencing scheme takes into account the directionality of information transport and is used to approximate the convective flux. The diffusive flux is approximated using linear interpolation ; we assume that T varies linearly between grid nodes. Upwind differencing is useful for high velocity profiles which match Péclet number that are are > 1. We also investigate false diffusion which is a drawback that comes with upwind differencing.

Problem statement

Given a 2 dimensional square domain of Length L = $1m$.
The Governing equation for Temperature reads :

$$\frac{\partial}{\partial x}(\rho u T) + \frac{\partial}{\partial y}(\rho v T) = \frac{\partial}{\partial x}(k\frac{\partial T}{\partial x}) + \frac{\partial}{\partial y}(k\frac{\partial T}{\partial y}) \tag{1}$$

where : $\rho = cst$, u, v are uniform on the domain and known , k is the thermal conductivity , T is the temperature.
Boundary conditions are given to be dirichlet boundary conditions :

- $T_{left} = 4$
- $T_{Top} = 1$
- $T_{right} = 0$
- $T_{bottom} = 0$

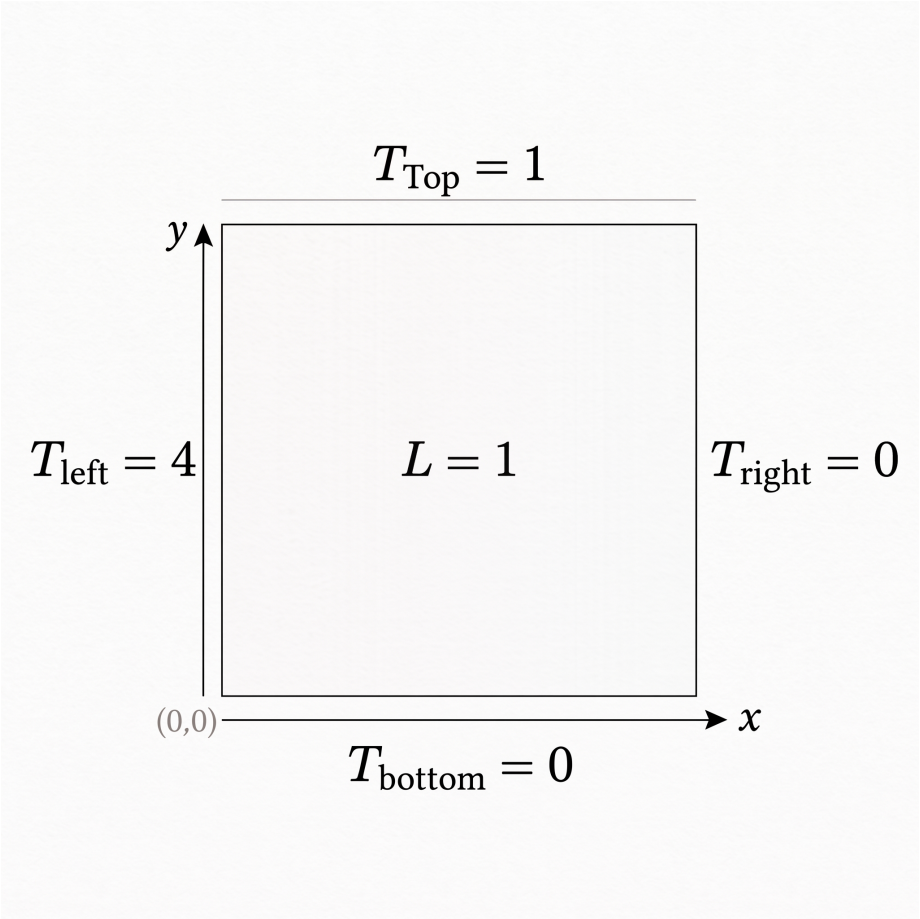


Figure 1: 2D square domain

Discretization

We want to have a discretization with uniform mesh i.e $h_x = h_y = h$
Vertex centered FVM is used to construct 2D grid of $N = 21$ x21 Nodes/Control volumes. Grid spacing $h = \frac{L}{N-1}$. Assuming that $A_e = A_w = A_n = A_s = A$. ie a uniform crossectional area of the meshes.

Integrate the governing equation (1) onto a finite volume centered at P and using Gauss theorem we obtain the flux equation:

$$F_e T_e - F_w T_w + F_n T_n - F_s T_s = k_e \frac{\partial}{\partial x} T_e - k_w \frac{\partial}{\partial x} T_w + k_n \frac{\partial}{\partial y} T_n - k_s \frac{\partial}{\partial y} T_s$$

(Flux equation)

Where:

- F = Convection strength =convective transport coefficient = $\rho u, \rho v$
- D = Deffusive strength =diffusive transport coefficient = k/h

$F_e = F_w = F_x$, $F_n = F_s = F_y$ due to conservativeness .

Since we have uniform meshes and k is uniform it implies that $D_e = D_w = D_n = D_s = D$

After considering all the 4 cases for the upwind scheme ; $u > 0, v > 0, u < 0, v > 0, u > 0, v < 0, u < 0, v < 0$ we arrive at the general form of the discretisation equation:

$$a_P T_P = a_E T_E + a_W T_W + a_N T_N + a_S T_S \tag{2}$$

with :

1. $a_W = D + max(0, F_x)$
2. $a_E = D + max(0, -F_x)$
3. $a_N = D + max(0, -F_y)$
4. $a_S = D + max(0, F_y)$

For boundedness of the scheme we require that :

$0 \leq \frac{a_W}{a_P} \leq 1$, $0 \leq \frac{a_E}{a_P} \leq 1$, $0 \leq \frac{a_N}{a_P} \leq 1$, $0 \leq \frac{a_S}{a_P} \leq 1$

Numerical Results

Figure 2 represents the convection and diffusion of Temeperature on the 2D grid for low velocities using Upwind scheme.

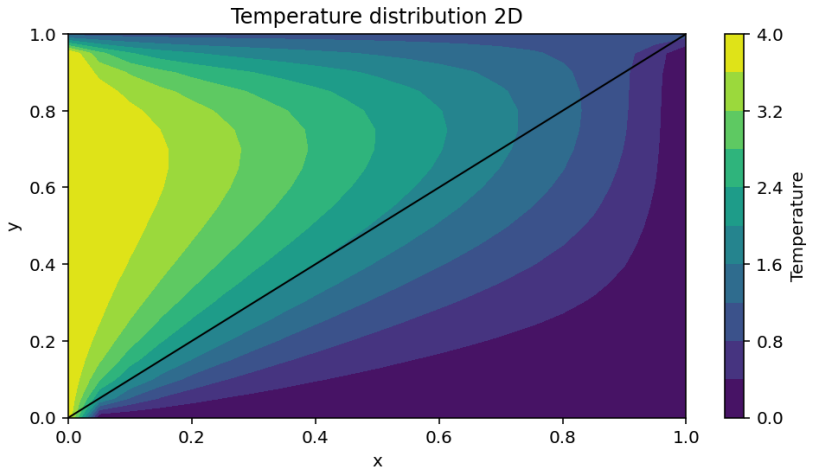


Figure 2: 2D Temperature distribution for low velocity profile

Figure 3 represents the convection and diffusion of Temeperature on the 2D grid for high velocities using Upwind scheme.

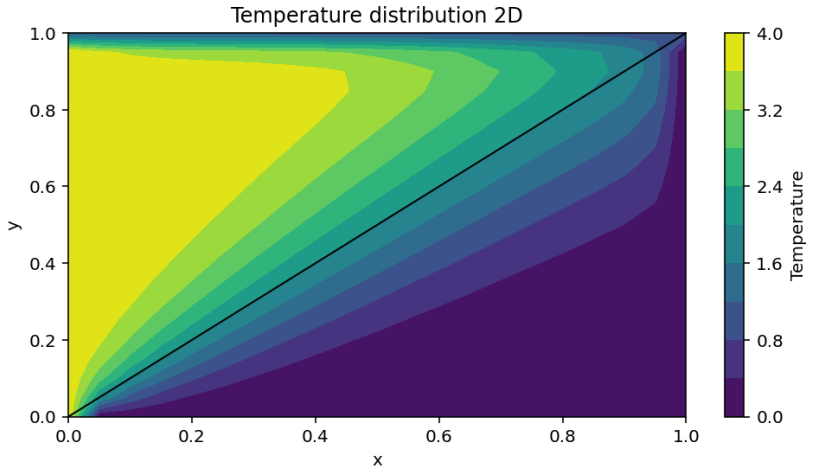


Figure 3: 2D Temperature distribution for high velocity profile

Figure 4 represents the convection of Temperature , for this simulation we turned off diffusion to see what happens.

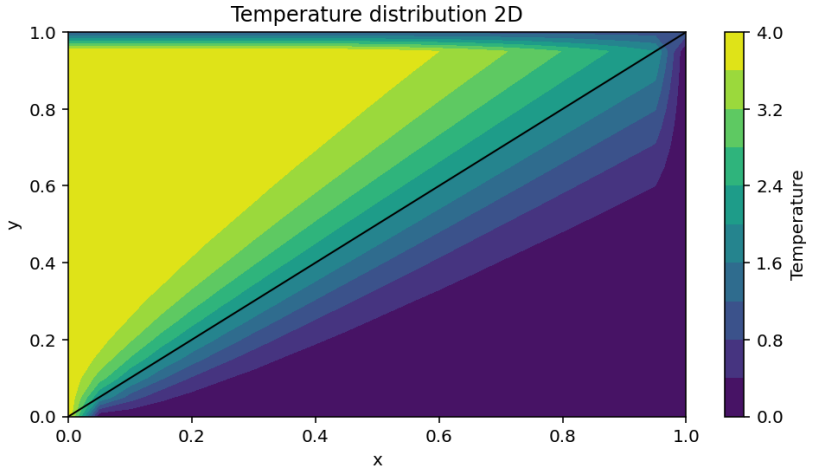


Figure 4: 2D Temperature distribution when diffusion is turn OFF

Discussion of Results

For the convection and diffusion of Temeperature on the 2D grid for :

1. for low velocity profile : u = v = 30
   Pe in x = 0.375 , Pe in y= 0.375
   Temperature distribution obtained after 876 iterations

2. for high velocity profile : u = v = 300
   Pe in x = 3.75 , Pe in y= 3.75
   Temperature distribution obtained after 132 iterations

First order Upwind differencing scheme deals perfeclty with low and high péclet number , it produces a bounded,conservative and accurate result. The downside of the upwind scheme can be seen on figure 4 which is called **False Diffusion**

False diffusion

False diffusion is actually a special case of the simulation when Diffusion is turned off. This is achieved by setting k = 0 , then the problem becomes a convection problem.

Here the velocity is diagonal to the cartesian plane , 45° above x axis (diagonal line shown in black on the figure). Because there is no diffusion the temperature is not supposed to spread and since the velocity is positive here , flow is from left to right meaning that only the temperature to the left side of the domain will be transported 45° above x axis along the diagonal $y = x$, the Top temperature will not be transported. Convection only transports the property , it does not change the value but here we can clearly see on Figure 4 near the diagonal line temperature value changes from 4 and decreases along the perpendicular direction to the diagonal. This suggests diffusion and it was not supposed to happen since we have set k (thermal conductivity) = 0, in such cases we call this False diffusion.